

Nirmal Kumar S

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PROFILE

Generative AI Engineer and Design Technologist with 9+ years delivering production-scale generative AI systems and creative automation tools at Amazon, DNEG, and Technicolor. Expert in LLM, agentic AI, full-stack development (Python, React, Django), and AWS cloud architecture. Proven track record shipping AI workflows for millions of users, achieving measurable conversion improvements, and boosting team efficiency 25% through pipeline innovation.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, SQL

Frameworks & Libraries: React JS, Django, PySide, PyQt, Tailwind CSS, PostgreSQL, REST API, FAST API, Langchain, Langgraph

AI Technologies: LLM, RAG, Agentic AI, MCP

Cloud (AWS): EC2, Lambda, S3, RDS, IAM, Cloudwatch, Cloudfront, SQS, SNS

Pipeline & Production Tools: Figma, USD, Ftrack, Deadline

Development Tools: GIT, Poetry, Slack API, FAL, Kling, Kontext, SeedVR

Domain Expertise: System Design, Pipeline Development, Software Engineering, Generative AI, Automation

EXPERIENCE

Design Technologist II, Amazon 04/2025 – Present | Chennai

- Collaborated with brand partner teams, program managers, and operations POCs to create Gen AI workflows for white elevated backgrounds, delivering consistent Premium CX image quality across major events and achieving a 3.12% conversion rate.
- Collaborated with SDE teams, program managers, UX designers and stakeholders to develop front end pages and launch high profile brand pages for Saks Fifth Avenue [🔗](#), Steelers [🔗](#) and Nike [🔗](#) for Amazon's platform.
- Designed and developed Workbench, an internal Gen AI-powered self-service platform enabling marketing teams to generate product images, lifestyle imagery, ghost mannequin visuals, and videos, while automating bulk content generation for Amazon's website.
- Spearheaded system design, and development of Editorial artist production pipeline, created artist facing tools using PySide and Python, thereby streamlining artist workflow, versioning, storage and improving overall artist efficiency by 20%.
- Built a Python microservice integrating Meta's Segment Anything Model (SAM) for automated image segmentation using a FAST API service reducing segmentation and overall inpainting effort by 40%.
- Developed an internal search tool powered by LLM technology that enables users to search the internal image library and retrieve visually similar images.
- Developed an agentic AI system using MCP to retrieve project information from Lambda function code.
- Mentored junior developers in software development practices, driving improved code quality and establishing team-wide coding standards.

Design Technologist I, Amazon 08/2021 – 03/2025 | Chennai

- Developed a cloud based rendering tools for Maya and Blender using Deadline cloud and created support for Arnold and V-Ray.
- Created automated headless batch rendering scripts that dynamically swap 3D objects and render them in specified backgrounds, enabling scalable asset generation on the render farm.
- Developed a comprehensive portal for task management, user attendance tracking and artist performance metrics monitoring which streamlined project oversight and improved management efficiency for project coordinators and team managers.
- Acted as Technical POC, partnering with third-party vendors including Adobe, Ftrack, and Smartsheet to test new features and products, evaluate content workflow improvements, and successfully integrate solutions into production pipelines.
- Ambassador for Amazon 3D Commerce in the Khronos Group.
- Mentored non-technical users in scripting fundamentals, enabling them to develop custom plugins and automation scripts that streamlined their team's workflows.

Pipeline ATD, DNEG 11/2020 – 08/2021 | Chennai

- Developed an artist-facing Maya tool for importing and managing light rigs, reducing lighting setup time and standardizing workflows across production.
- Contributed to the development of a centralized asset library for 3D and 2D assets, fostering reusability and reducing redundant asset creation across teams.
- Provided technical support and resolved pipeline-related tickets by troubleshooting system issues, implementing bug fixes, and ensuring seamless production workflows.

Software Analyst, Technicolor

05/2017 – 11/2020 | Bengaluru

- Developed automated sync tools to replicate and monitor data across multiple studio sites.
- Automated render farm operations by developing Python scripts for headless batch rendering reducing manual intervention in render workflows.
- Managed studio software environments by maintaining validated tool chains and coordinating software deployments across artist workstations.
- Proactively identified technical issues and resolved support tickets on rotation, implementing bug fixes to maintain pipeline stability

EDUCATION

M.Tech, Software Engineering, Birla Institute of Technology and Science
First Class, CGPA: 7.5/10

2021 – 2022 | Pilani

B.Tech, Civil Engineering, SASTRA University
First Class, CGPA: 7.1/10

2012 – 2016 | Thanjavur